

- 6 (a) difference: vibration / oscillation (of particles) / displacement of particles is parallel to energy transfer / wavefronts in longitudinal and perpendicular for transverse B1
 or
 transverse can be polarised, longitudinal cannot be polarised
- similarity: both transfer / propagate energy B1 [2]
- (b) (i) waves from slits are coherent / constant phase relationship (B1)
 waves overlap (at screen) with a phase difference or have a path difference (B1)
 maxima where phase difference is integer $\times 360^\circ$ (or $\times 2\pi$ rad)
 or path difference is integer $\times \lambda$
 or equivalent explanation of minima e.g. $(n + \frac{1}{2}) \times 360^\circ$ (B1)
 max. 2 [2]
- (ii) maxima spacing = $\lambda D / a$ C1
 = $(6.3 \times 10^{-7} \times 2.5) / 0.35 \times 10^{-3}$
 = $4.5 \times 10^{-3} \text{ m}$ A1 [2]
- (c) (ultra-violet has) shorter wavelength, hence smaller separation / distance A1 [1]